



Essay / Assignment Title:

Transforming Retail Data into Strategic Business Insights using Tableau

Programme title: MSc Data Analytics

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CONTENTS

INTRODUCTION	4
CHAPTER ONE: DATASET, PREPARATION AND TABLEAU DESIGN	5
1.1 Business problem, stakeholders and analytical questions	5
1.2 Dataset source, structure and reliability	5
1.3 Data preparation, transformation, ethics and limitations	5
1.4 Tableau implementation and visual design	6
CHAPTER TWO: DASHBOARD ANALYSIS AND FINDINGS	7
2.1 Executive Overview Dashboard	7
2.2 Customer and Product Performance Dashboard	8
2.3 Regional and Operational Insights Dashboard	9
CHAPTER THREE: BUSINESS VALUE, RECOMMENDATIONS AND CRITICAL EVALUATION	11
3.1 Recommendations	11
3.2 Critical evaluation and future analysis	11
CONCLUDING REMARKS	12
BIBLIOGRAPHY	13
APPENDIX	14

Supporting project repository: https://github.com/rrioroymanjaly-ui/BSBI_Data_Analytics_Assignment

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INTRODUCTION

Retail data can look positive at first glance when sales are increasing, but sales alone do not show whether the business is actually improving. A category can sell well and still return very little profit, while a smaller segment can be more efficient. I used Tableau to look at these differences together instead of treating revenue, customers, products and regions as separate topics. The aim was to build a set of views that a manager could use to understand what happened, investigate the reason behind it and decide where attention is needed.

Before building the Tableau views, I wrote down the questions a manager would actually ask. Is the business growing in both Sales and Profit? Which products and categories make money rather than only revenue? Which customer segments are worth the most? Are there seasonal or geographical patterns? Where do discounting, shipping or regional results suggest a problem? I built the dashboards for senior managers, sales and marketing teams, and regional managers, starting with the overall position before moving into the detail each team needs.

I used the Orders table from Tableau Sample - Superstore. Tableau describes Superstore as a sample source with product, customer and sales information for a fictitious retailer (Tableau, 2026a). The commercial fields needed for this project are available in one transactional dataset. I organised the analysis into three dashboards: Executive Overview, Customer and Product Performance, and Regional and Operational Insights.

Superstore is a teaching dataset, so the results demonstrate an analytical method rather than describe a real company. I also avoided turning correlations into causal claims. The data, for example, show weaker profitability at deeper discounts, but they do not prove that discounting alone caused each loss. That distinction matters when moving from a visual pattern to a business recommendation (Munzner, 2014).

CHAPTER ONE: DATASET, PREPARATION AND TABLEAU DESIGN

1.1 Business problem, stakeholders and analytical questions

The main business problem is that a large transaction file does not automatically give management a clear picture of performance. Looking only at Sales would favour the biggest categories, customers and regions, even if they generate poor margins. I therefore compared Sales and Profit with Profit Ratio, order value, discounting and the number of distinct orders. This made it possible to separate growth from profitable growth and to move from an overall result to the product, customer or location behind it.

1.2 Dataset source, structure and reliability

The Orders table contains 9,994 product-level rows from 2014 to 2017, representing 5,009 distinct orders and 793 customers. One order can contain several products, so row count is not the same as order count. The file also contains four regions, 49 states, three categories and 17 sub-categories. Order-based measures therefore use COUNTD([Order ID]) rather than counting rows.

For the analysis I treated Order Date, Ship Date, Ship Mode, Segment, Region, State, City, Category, Sub-Category and Product Name as dimensions. Sales, Profit, Quantity and Discount were used as measures. Order ID and Customer ID were kept as identifiers for distinct counts, while Postal Code was used only as a geographical field. Setting the field roles correctly was important because an incorrect aggregation can produce a convincing chart with the wrong answer.

Before building the dashboards, I checked the basic data quality and control totals. The file has 21 fields, no missing values in the fields used here and no duplicate Row IDs. Total Sales reconcile to \$2,297,200.86 and total Profit to \$286,397.02. I used these figures to check later calculations. The sample is internally consistent, but it is not representative evidence for the wider retail industry (Tableau, 2026a).

Validation item	Result	Analytical implication
Rows	9,994	Order-line grain
Distinct orders	5,009	Use COUNTD(Order ID)
Distinct customers	793	Use COUNTD(Customer ID)
Date coverage	2014-2017	Four-year trend analysis
Regions / categories	4 / 3	Regional and portfolio comparison
Missing values in analysis fields	0	No imputation required
Total Sales / Profit	\$2,297,200.86 / \$286,397.02	Baseline reconciliation

Table 1. Dataset profile and validation checks.

1.3 Data preparation, transformation, ethics and limitations

Only limited cleaning was needed because the source is already structured. I checked date and numeric field types and reviewed Category, Region and Ship Mode labels for consistency. Negative-profit rows and large orders were kept because they are valid business observations, not errors. Removing unusual but genuine transactions would make the dashboard cleaner while hiding information that management may need to investigate.

I created three drill-down hierarchies: Time (Year > Quarter > Month > Order Date), Product (Category > Sub-Category > Product Name) and Geography (Country > Region > State > City). Tableau hierarchies allow the user to move from a summary into detail without creating a new chart for every level (Tableau, 2026d). I also grouped shipping modes for one comparison and created Discount Bands for 0%, low (>0-10%), medium (>10-30%) and high (>30%).

Privacy risk is limited because Superstore is fictitious, but I still kept customer analysis at commercial segment level and did not infer personal characteristics from names. A low-profit state is treated as a reason to investigate, not a reason to reduce service automatically. The dataset also lacks inventory, detailed product cost, marketing spend, competitor activity and service-quality measures, which limits causal explanation and more advanced customer or

logistics analysis. The brief describes a multi-channel retailer, but this sample does not contain a sales-channel field, so I do not invent or infer channel-level performance.

1.4 Tableau implementation and visual design

Several calculated fields were added because the raw columns do not answer all of the management questions directly. Profit Ratio is $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$, which shows how much profit is retained from each dollar of sales. Average Order Value is $\text{SUM}([\text{Sales}]) / \text{COUNTD}([\text{Order ID}])$. Shipping Days uses $\text{DATEDIFF}(\text{'day'}, [\text{Order Date}], [\text{Ship Date}])$, and Loss Flag separates negative-profit results from profitable ones. DATEPART and DATETRUNC were used for time analysis, together with a year-on-year percentage change on the annual trend.

Two parameters make the workbook easier to explore. Metric Selector switches a view between Sales, Profit and Profit Ratio, while Top N changes the number of products or customers displayed (Tableau, 2026b). Filter actions connect related charts, so clicking a region, state, category or segment narrows the other views and allows an exception to be investigated in context (Tableau, 2026c).

I chose chart types according to the question. Time is shown with line charts, rankings with horizontal bars, geography with a map linked to supporting views, and Discount versus Profit with a scatter plot and linear trend line (Tableau, 2026e). I did not forecast beyond 2017 because projecting a fictitious historical sample would create false precision.

The layout is deliberately simple. KPI cards come first, filters stay in consistent positions and extra detail is placed in tooltips. Colour separates positive and negative performance rather than decorating the page. This follows guidance from Few (2013), Munzner (2014) and Knaflic (2015) on clarity, task-focused visual choice and directing attention to the main message.

CHAPTER TWO: DASHBOARD ANALYSIS AND FINDINGS

2.1 Executive Overview Dashboard

The Executive Overview gives the first answer to whether the business is growing and whether that growth is profitable. Across the four years, the dataset records \$2,297,200.86 in Sales and \$286,397.02 in Profit. This gives an overall Profit Ratio of 12.47%. There are 5,009 distinct orders and 793 customers, and Average Order Value is about \$458.61. I placed these measures together because a large sales figure can be misleading if the margin behind it is weak.

The yearly pattern is not a straight line. Sales are about \$484.2k in 2014, dip to \$470.5k in 2015, then rise to around \$609.2k in 2016 and \$733.2k in 2017. Profit increases from about \$49.5k in 2014 to \$93.4k in 2017. Sales therefore grow by roughly 51% from 2014 to 2017. Monthly results are stronger later in the year, but the Year, Category and Region filters are needed before using that seasonal pattern for inventory or campaign decisions.

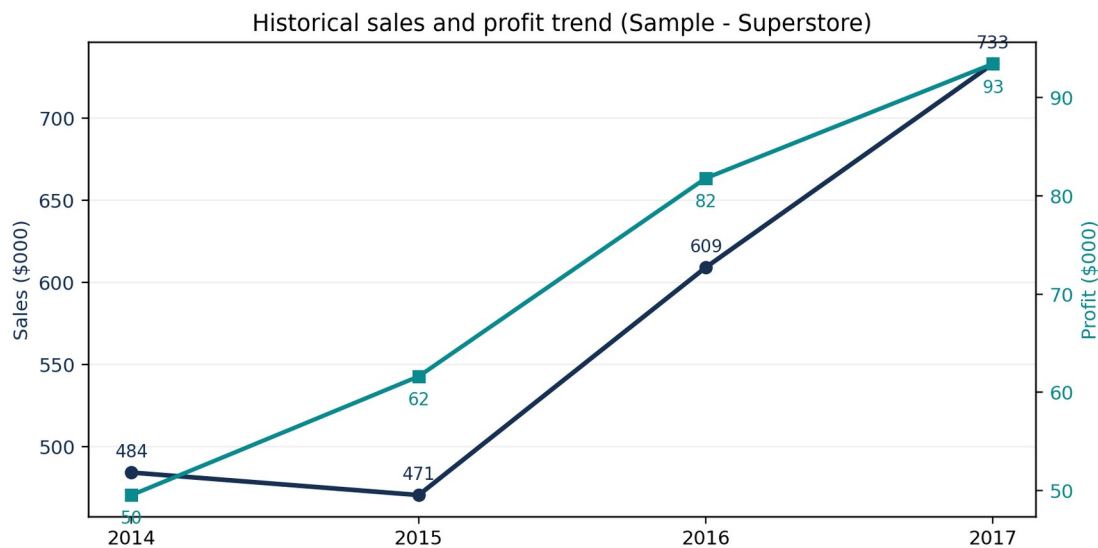


Figure 1. Historical sales and profit trend, 2014-2017. Source: author analysis of Sample - Superstore.

West is the strongest region overall, with about \$725.5k Sales and \$108.4k Profit, a Profit Ratio near 14.9%. East follows at roughly \$678.8k Sales, \$91.5k Profit and 13.5%. Central produces around \$501.2k Sales but only \$39.7k Profit, giving about 7.9%, while South records around \$391.7k Sales and \$46.7k Profit. Central therefore has a meaningful sales base but converts it into profit less efficiently than the other regions.

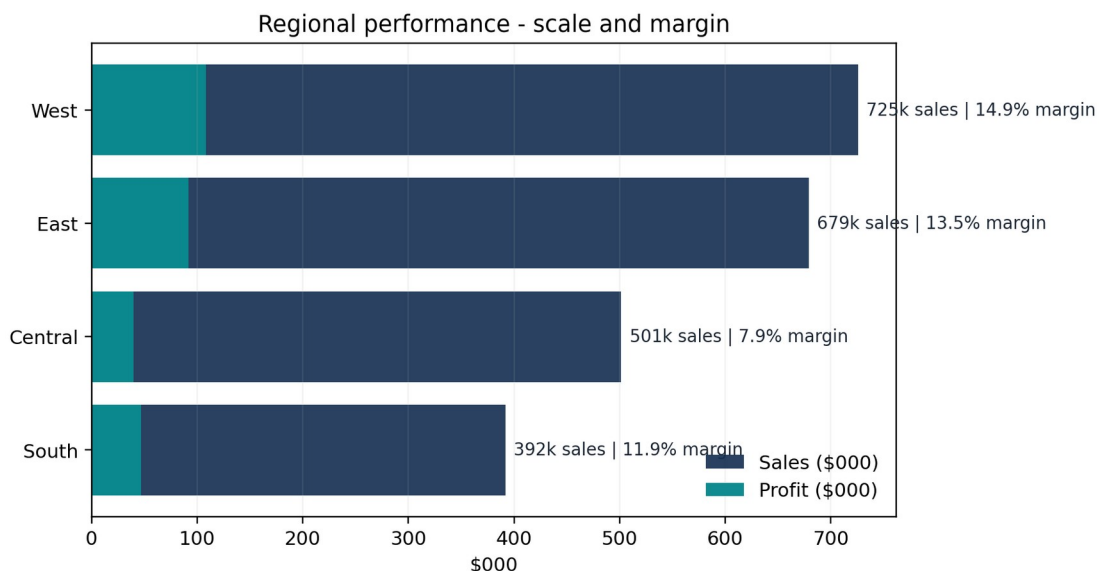


Figure 2. Regional sales, profit and profit-ratio comparison.

2.2 Customer and Product Performance Dashboard

Customer value changes depending on the measure. Consumer is largest, with about \$1.161m Sales and \$134.1k Profit. Corporate contributes roughly \$706.1k Sales and \$92.0k Profit, while Home Office contributes about \$429.7k Sales and \$60.3k Profit. Consumer is the most valuable segment in absolute terms, but Home Office has the best Profit Ratio at about 14.0%, ahead of Corporate at around 13.0% and Consumer at 11.5%.

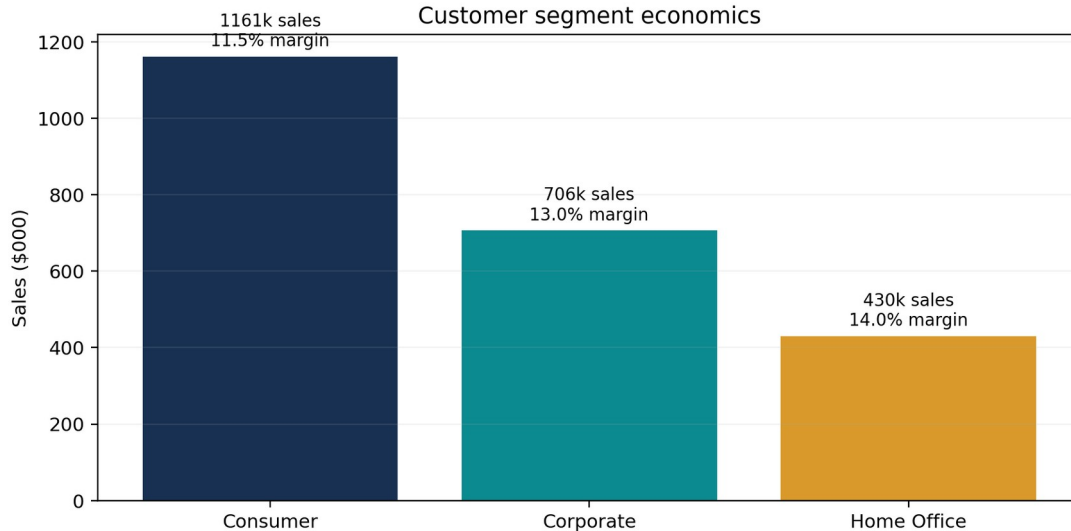


Figure 3. Customer segment economics: sales scale and profit ratio.

The category comparison gives an even clearer example. Technology produces about \$836.2k in Sales and \$145.5k in Profit, which is a Profit Ratio close to 17.4%. Office Supplies records roughly \$719.0k Sales and \$122.5k Profit, around 17.0%. Furniture has about \$742.0k Sales, so it looks strong on revenue, but Profit is only around \$18.5k and the Profit Ratio is close to 2.5%. If the dashboard showed only Sales, Furniture would appear healthier than Office Supplies even though its economics are much weaker.

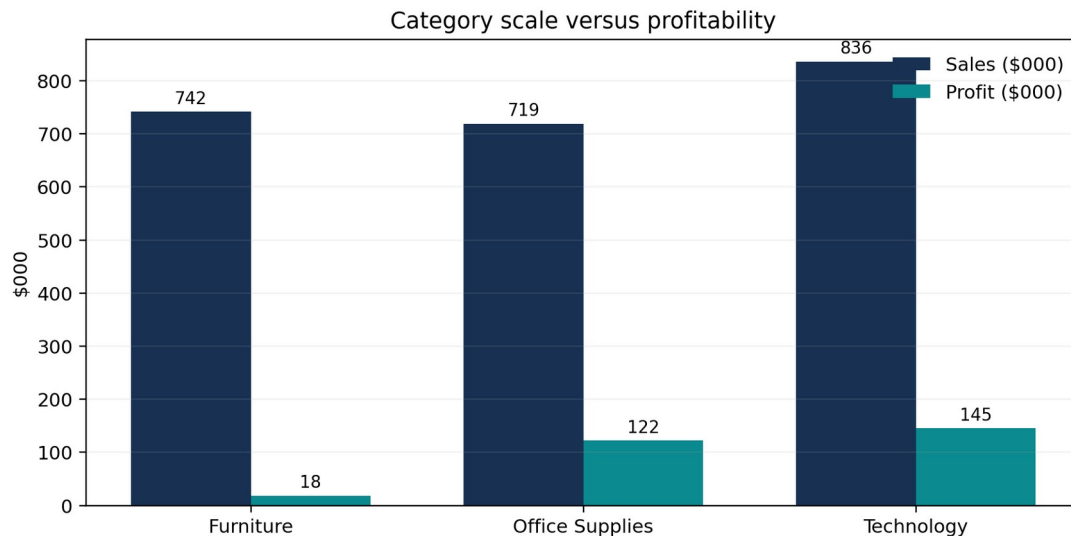


Figure 4. Category scale versus profitability.

The sub-category view explains the category gap. Copiers generate about \$55.6k Profit, Phones \$44.5k and Accessories \$41.9k. Tables lose roughly \$17.7k, Bookcases \$3.5k and Supplies \$1.2k. Machines also deserve attention because Sales are meaningful but Profit is modest. Canon imageCLASS 2200 Advanced Copier is the leading product in the standard dataset, with about \$61.6k Sales and the highest product-level Profit. Top N lets the user move from portfolio results to individual products or customers.

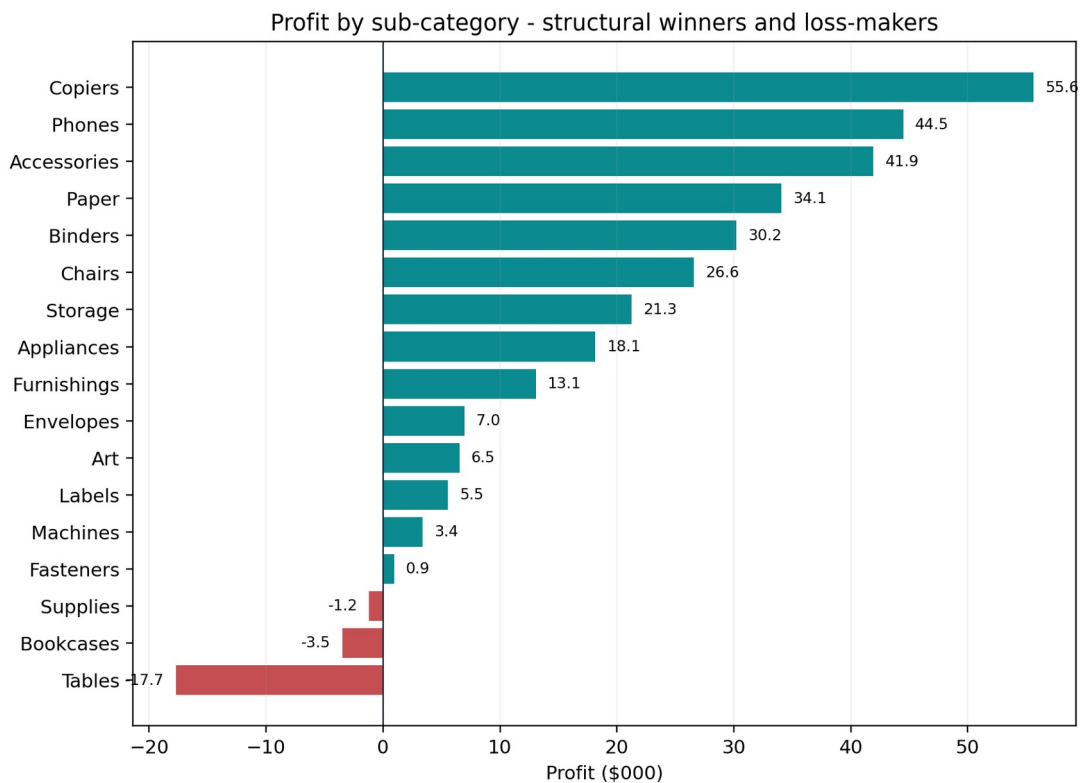


Figure 5. Profit by sub-category, highlighting structural winners and loss-makers.

For me, the useful conclusion from this dashboard is not "sell more of the biggest category". A large category deserves more investment only when the margin remains healthy. The same applies to customer segments: smaller segments can still be attractive if they convert Sales into Profit efficiently. Using Metric Selector to move between Sales, Profit and Profit Ratio keeps these three views of performance in the same filter context.

2.3 Regional and Operational Insights Dashboard

The state view adds detail to the regional result. California and New York are the main positive profit contributors, while Texas stands out as a loss-making state in the standard dataset. I linked the map to Category, Segment and Discount Band so that selecting a state shows whether the problem is spread across the business or concentrated in a particular product or customer group. This makes the map useful for investigation rather than simply showing where sales happened.

Shipping Mode adds context, but the data do not support a strong claim about shipping efficiency. Standard Class represents about 59% of Sales and approximately \$164.1k Profit. Shipping Days can be calculated from the order and ship dates, but actual fulfilment cost, delays and customer satisfaction are missing. I therefore use Ship Mode and Shipping Days as supporting filters rather than claiming that one method is automatically better.

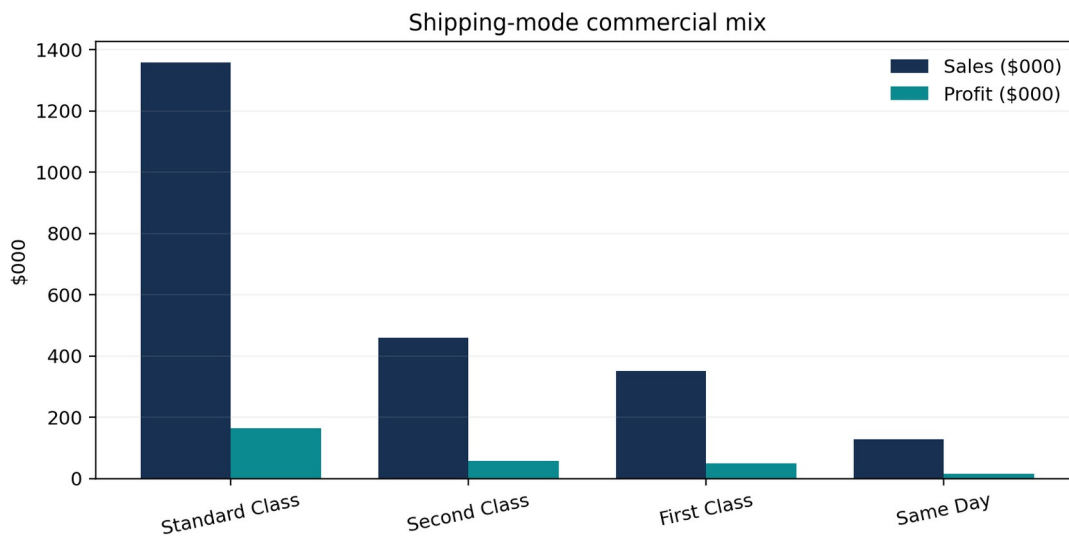


Figure 6. Sales and profit by shipping mode.

Discounting gives a clearer operational warning. In the scatter view, profitability becomes weaker as discount depth increases and the linear trend line slopes downward. This is an association, not proof of a direct causal effect, because product mix and other factors are also involved. Even so, the pattern becomes more important when it is combined with the product results. Tables and Bookcases already lose money, so high-discount transactions in those sub-categories deserve immediate review. Central should also be checked by product mix and discount band because its regional margin is already well below West and East.

CHAPTER THREE: BUSINESS VALUE, RECOMMENDATIONS AND CRITICAL EVALUATION

3.1 Recommendations

The first priority is to protect profitable growth in Technology. It has the strongest combination of Sales and Profit, and Copiers, Phones and Accessories are important contributors. I would use the dashboard to identify products and customer groups where Technology sales are growing without a drop in Profit Ratio. Promotional activity should be judged on both revenue and profit rather than on volume alone.

Furniture needs a different response. Its sales base of about \$742.0k is too large to ignore, but a Profit Ratio close to 2.5% shows that the category is not converting revenue effectively. Tables and Bookcases should be reviewed by product, state, segment and discount level. The objective is not to stop selling Furniture; it is to find the combinations where discounting or product mix repeatedly produces losses and correct those first.

Central is the main regional area I would investigate next. Its Profit Ratio of around 7.9% is far below West and East even though the region has more than \$500k in Sales. Before changing targets or moving resources, I would use the state, category, segment and Discount Band filters to see whether the gap is driven by a few loss-making combinations or is more widespread.

The late-year demand pattern can also support planning, but it should be used carefully. Instead of raising stock across every category before Q4, managers can drill through the Time and Product hierarchies to identify the category-region combinations where demand rises and Profit Ratio remains healthy. This would make inventory and campaign decisions more focused and reduce the risk of increasing low-margin sales.

Finally, I would use the workbook as a regular management view rather than only for this project. A monthly review could track Sales, Profit, Profit Ratio, year-on-year change, loss-making sub-categories, weak regions and high-discount activity using the same definitions. The parameters and filter actions let different managers explore their own area without changing the underlying calculations, which is important if the dashboard is used by several teams.

3.2 Critical evaluation and future analysis

The strongest part of the approach is that the measures are connected. Orders are counted distinctly, Sales are not treated as a substitute for Profit, and the geographical and product views can be filtered together. The dashboard also keeps the discount result in proportion: the negative relationship is visible, but I do not treat the trend line as proof of causation. This matters because visualisations can make a weak assumption look more certain than it really is.

The biggest limitation remains the source itself. Sample - Superstore is a fictitious teaching dataset, so this report demonstrates how I would analyse a retail business rather than making a claim about a real retailer's future. The file also does not contain acquisition cost, inventory levels, marketing spend, complete returns information, fulfilment cost, service failures or competitor data. As a result, contribution margin, campaign ROI, stock-out effects, customer lifetime value and the real cost of faster shipping cannot be calculated reliably from this dataset alone.

In a real business, I would connect the dashboard to governed live data and keep the same reconciliation checks on every refresh. Product cost and returns would improve the margin view; inventory would reveal stock-outs; marketing data would allow campaign ROI; and service data would strengthen shipping analysis. Forecasting would then be useful once a genuine time series could be tested on a holdout period.

CONCLUDING REMARKS

The results show why retail performance needs both scale and profitability. Sales improve strongly after 2015 and peak in 2017. West produces the highest regional Profit, Consumer leads in absolute segment contribution, and Technology is the strongest category when Sales and margin are considered together. Home Office is smaller but has the best segment Profit Ratio. The main weak areas are Furniture's low margin, losses in Tables and Bookcases, a notable loss position in Texas, and Central's weak conversion of Sales into Profit.

Tableau makes these results easier to use because the reader can move from the overall numbers to the reason behind them. The Time, Product and Geography hierarchies support drill-down; Metric Selector and Top N change the question without rebuilding a chart; calculated fields add Profit Ratio, order value, shipping time and loss indicators; and filter actions connect the dashboards. Together, these features allow managers to move from overall performance to specific exceptions and then to the areas that need action.

Based on the evidence in the dataset, I would protect profitable Technology growth, review loss-making Furniture lines, investigate Central and Texas in more detail, and plan late-year activity only where higher demand is accompanied by healthy margin. I would also monitor discounting closely because the weakest product results and deeper discounts appear together often enough to justify further investigation. These recommendations are deliberately limited to what the data can support. In a real retail environment, the same dashboard structure would become much more valuable when combined with cost, inventory, marketing, returns and service data.

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APPENDIX

Appendix A: Data dictionary

Field	Type / role	Use in analysis
Order Date	Date / dimension	Time trend, hierarchy, year/month filters
Ship Date	Date / dimension	Shipping Days calculation
Order ID	Identifier / dimension	Distinct order count and AOV
Customer ID	Identifier / dimension	Distinct customer count and rankings
Segment	Categorical dimension	Customer segment comparison
Region / State / City	Geographic dimensions	Regional and state drill-down
Category / Sub-Category / Product Name	Product dimensions	Product hierarchy and Top N
Ship Mode	Categorical dimension	Operational comparison and group
Sales	Continuous measure	Revenue scale
Profit	Continuous measure	Profitability
Quantity	Whole-number measure	Volume context
Discount	Continuous measure	Discount-band and trend analysis

Appendix B: Calculated fields, parameters, groups and hierarchies

Item	Tableau definition / configuration
Profit Ratio	SUM([Profit]) / SUM([Sales])
Average Order Value	SUM([Sales]) / COUNTD([Order ID])
Shipping Days	DATEDIFF('day',[Order Date],[Ship Date])
Loss Flag	IF SUM([Profit]) < 0 THEN 'Loss' ELSE 'Profit' END
Discount Band	0%; Low (>0-10%); Medium (>10-30%); High (>30%)
Metric Selector parameter	Sales / Profit / Profit Ratio
Top N parameter	User-controlled number of customers/products displayed
Ship Mode Group	Expedited: Same Day + First Class; Standard/Economy: Second + Standard Class
Time hierarchy	Year > Quarter > Month > Order Date
Product hierarchy	Category > Sub-Category > Product Name
Geography hierarchy	Country > Region > State > City

Appendix C: Dashboard layouts

Executive Overview Dashboard

Filters: Year | Region | Segment | Category

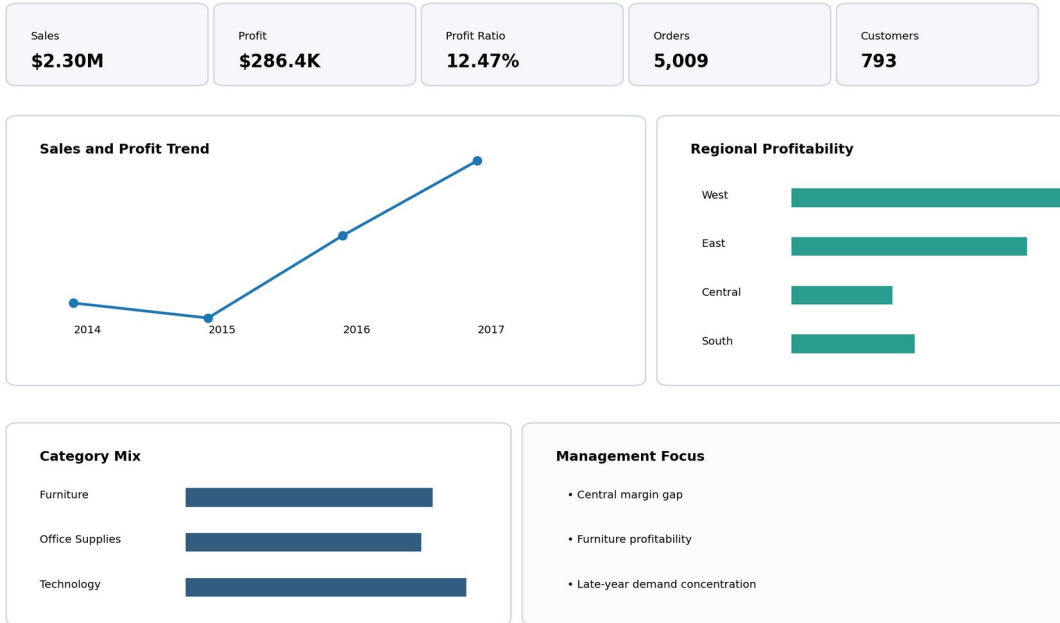


Figure A1. Executive Overview Dashboard layout.

Customer and Product Performance Dashboard

Parameters: Metric Selector | Top N Filters: Year | Region | Category

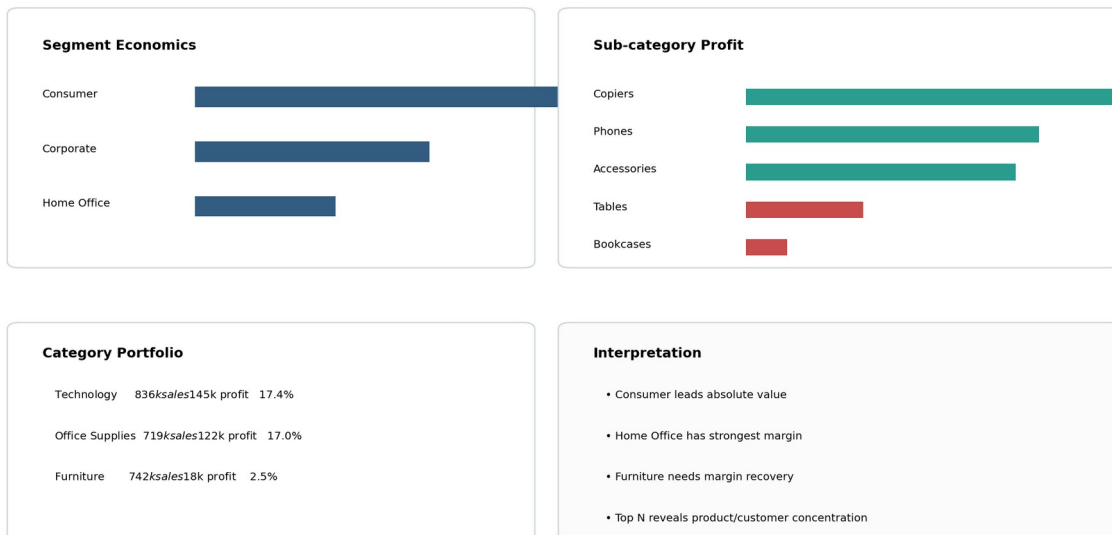


Figure A2. Customer and Product Performance Dashboard layout.

Regional and Operational Insights Dashboard

Filters: Region | State | Category | Segment | Discount Band | Ship Mode

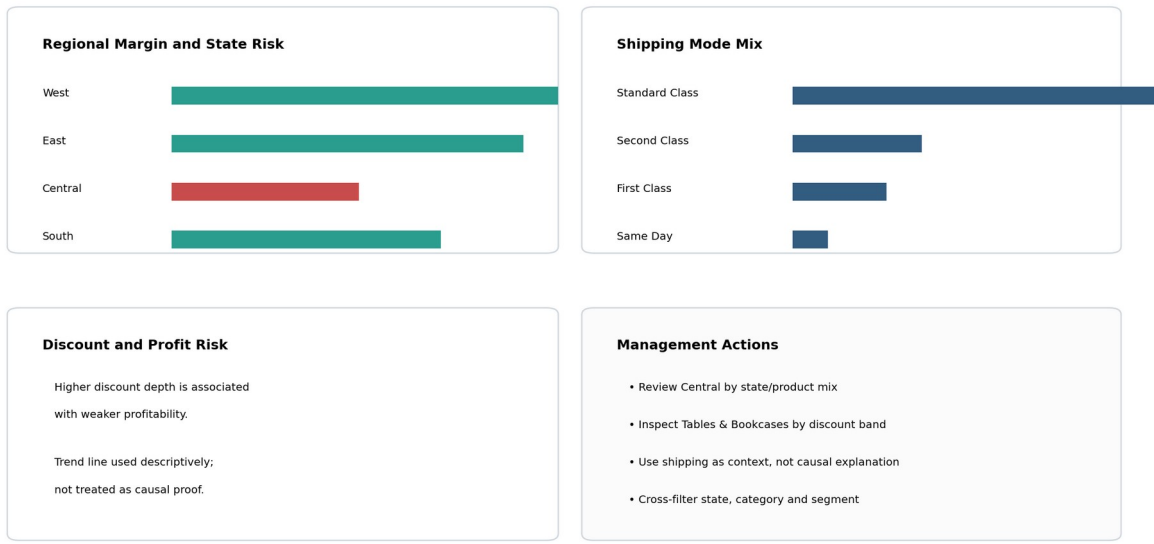


Figure A3. Regional and Operational Insights Dashboard layout.

Appendix D: Supporting project repository

GitHub repository: https://github.com/rrioroymanjaly-ui/BSBI_Data_Analytics_Assignment